

Whitening and sparse flow-matching models

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Completed follow-up on synthetic multivariate GP, ETTh1 and Exchange. Every neural model is evaluated with both a temporally white source and a smooth GP source. Sparse autoencoders condition the flow-matching model on history; they are not scored as standalone predictors.

Completed evidence	Scope
288 new MPS fits	264 main fits plus a 24-fit dictionary-rank follow-up
96 frozen reference fits	MLP, fixed-mask MLP, S4 and U-Net; same data, seeds and scoring
New linear control	MultiTask Elastic Net; validation-selected regularization
Verification	384 fits reviewed including controls; independent audits and 36 CPU/MPS parity checks

Does joint PCA whitening help MLP?

Dataset	Source	Raw MLP CRPS	Whitened CRPS	CRPS change	Residual MSE change
GP (3 channels)	white	0.3288	0.3439	+4.6%	-6.6%
GP (3 channels)	gp	0.3328	0.3095	-7.0%	-14.3%
ETTh1 (OT only)	white	0.1422	0.1284	-9.7%	-5.1%
ETTh1 (OT only)	gp	0.195	0.1321	-32.3%	-14.5%
ETTh1 (7 channels)	white	0.3621	0.5268	+45.5%	+51.4%
ETTh1 (7 channels)	gp	0.375	0.5298	+41.3%	+64.4%
Exchange (8 channels)	white	0.3137	0.7071	+125.4%	+124.0%
Exchange (8 channels)	gp	0.3852	0.564	+46.4%	+38.4%

Whitening lowers mean CRPS in 3/8 source/dataset cells and raw residual velocity MSE in 4/8. Whitening worsens mean CRPS under both sources on both real multivariate targets. These are distinct outcomes: improving velocity regression does not automatically improve the generated forecast distribution. Negative changes favor whitening.

Baseline check. Within the main 15-head comparison, the best simple baseline has lower CRPS than the lowest mean neural cell on 4/4 datasets. A favorable sparse-versus-dense ranking is therefore not, by itself, evidence of a useful forecasting improvement.

Most promising sparse neural result. The Top-2 mixture improves Exchange CRPS to 0.189 (white) and 0.235 (GP), versus 0.314 and 0.385 for MLP. It also improves on the dense mixture there. TopK history reconstruction and TopK velocity dictionaries do not show a comparable advantage over their dense controls at this budget.

Winner counts are descriptive, with no multiplicity correction. Paired context intervals condition on the selected periods and trained models; three seeds and two real datasets cannot establish a universal architecture claim.

What each comparison tests

Variant / control	Change inside conditional flow matching
Whitened MLP / raw MLP	Full joint time-channel PCA transform of state and velocity; inverse-mapped outputs; variance-weighted objective. History stays in original standardized coordinates.
Balanced-loss MLP / raw MLP	Only the training/selection loss is weighted by inverse PCA variances; state and head outputs remain in original coordinates.
TopK history SAE / dense AE	History reconstruction plus a 64-dimensional conditioning code; TopK retains at most 8 nonnegative features. Dense control uses the same encoder and decoder.
Sparse / dense conv. history	Three tied convolutional coding steps; sparse variant adds soft thresholds and code L1 loss. Both reconstruct observed history and condition the same FM architecture.
TopK / dense velocity dictionary	Predict a signed coefficient vector and combine learned velocity templates; TopK keeps at most 8 of 64 coefficients. No autoencoder reconstruction objective.
L0-gated / ordinary U-Net	Learn hard-concrete hidden-channel gates. Stochastic training, deterministic evaluation; measured gate activity distinguishes attenuation from actual pruning.
Top-2 / dense mixture	Four small velocity experts; sparse routing uses two. Both receive the same differentiable importance-balancing penalty.
Elastic Net / ridge	Linear multivariate forecasting with shared sparse selection of history lag/channel features; whole-vector training-residual bootstrap supplies samples.

Training and held-out evaluation

Approximately 64k allocated trainable parameters per new family, with the same width inside each dense/sparse pair. The 64-feature history AEs compress GP/ETTh1-seven-channel/Exchange histories; only the OT history has fewer than 64 inputs. Their dense controls share this bottleneck. The fixed-mask MLP reference matches active parameters and therefore allocates more weights. MLP whitening controls reuse the reference MLP width. Gates and thresholds add a small parameter overhead.

Each fit uses AdamW, learning rate 0.001, batch 128, 2,000 updates and three seeds (31, 32, 33). Validation is checked every 250 updates. Selection uses held-out CFM objective, excluding auxiliary penalties; whitened and balanced-loss models use their variance-weighted objective. Rates and regularization strengths are not tuned separately by architecture.

All variants receive the same observed history. Train-only global channel statistics normalize data. Real windows lie fully within chronological train/validation/test splits. Both sources have the same train-estimated cross-channel covariance; only temporal covariance differs. PCA is fitted separately by source to training velocities, with all $F \times C$ coordinates retained.

Evaluation uses 64 identical held-out contexts, 32 futures per context and 64 Euler steps. Fair CRPS scores marginal distributions; joint energy scores the complete time-channel vector, divided by $\sqrt{F \times C}$. Roughness is the mean absolute adjacent temporal increment relative to the target. Lower CRPS/energy is better; roughness near 1 alone does not imply calibrated or accurate forecasts.

Correlated GP: three channels

Head	White CRPS ↓	GP CRPS ↓	White energy ↓	GP energy ↓	White roughness	GP roughness
MLP	0.3288 ± 0.0037	0.3328 ± 0.0096	0.4301 ± 0.005	0.4309 ± 0.0099	1.12 ± 0.017	0.9809 ± 0.019
Whitened MLP	0.3439 ± 0.01	0.3095 ± 0.0018	0.4417 ± 0.0088	0.406 ± 0.0019	1.072 ± 0.019	0.9329 ± 0.013
Balanced-loss MLP	0.382 ± 0.0048	0.6157 ± 0.00029	0.4837 ± 0.0049	0.7085 ± 0.00042	1.211 ± 0.043	0.9188 ± 0.0072
Fixed-mask MLP	0.3257 ± 0.0072	0.3318 ± 0.0021	0.4272 ± 0.0076	0.4312 ± 0.0038	1.181 ± 0.033	1.203 ± 0.023
S4	0.333 ± 0.012	0.3318 ± 0.011	0.4357 ± 0.014	0.4297 ± 0.01	1.037 ± 0.031	0.9492 ± 0.05
U-Net	0.3215 ± 0.0058	0.3353 ± 0.012	0.4216 ± 0.0069	0.4336 ± 0.013	0.9535 ± 0.004	1.002 ± 0.051
L0-gated U-Net	0.4017 ± 0.023	0.3932 ± 0.026	0.4932 ± 0.023	0.4811 ± 0.025	4.518 ± 1.8	1.018 ± 0.033
Dense history AE	0.3391 ± 0.0061	0.3442 ± 0.0035	0.4397 ± 0.008	0.4414 ± 0.0027	1.679 ± 0.15	0.9825 ± 0.015
TopK history SAE	0.3897 ± 0.0047	0.3799 ± 0.0078	0.4872 ± 0.002	0.4751 ± 0.0066	1.658 ± 0.15	0.9712 ± 0.04
Dense conv. history	0.4033 ± 0.0069	0.3851 ± 0.0076	0.5015 ± 0.0054	0.4818 ± 0.0076	1.432 ± 0.047	0.9633 ± 0.038
Sparse conv. history	0.4075 ± 0.0043	0.3909 ± 0.0058	0.5059 ± 0.0029	0.4882 ± 0.0058	1.447 ± 0.023	0.9566 ± 0.041
Dense velocity dictionary	0.3243 ± 0.004	0.3329 ± 0.0053	0.4262 ± 0.0043	0.4308 ± 0.0049	1.316 ± 0.13	1.009 ± 0.024
TopK velocity dictionary	0.3815 ± 0.0073	0.3446 ± 0.0057	0.4741 ± 0.0091	0.4434 ± 0.0055	6.189 ± 0.07	1.329 ± 0.026
Dense mixture	0.3378 ± 0.0035	0.3368 ± 0.0075	0.437 ± 0.0037	0.4363 ± 0.0092	2.642 ± 0.055	0.9629 ± 0.018
Top-2 mixture	0.3391 ± 0.0032	0.3351 ± 0.0041	0.4382 ± 0.0035	0.4346 ± 0.0048	2.947 ± 0.23	0.9646 ± 0.026

Mean on the first line; ± sample SD across three optimization seeds on the second. All scores use train-standardized units. Compare methods within this dataset. White denotes temporally white noise with cross-channel covariance retained.

Simple forecasting controls

Baseline	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Persistence	0.7182	0.9072	0.9753	0.0%
Ridge + residuals	0.1929	0.2806	0.2039	86.4%
Elastic Net + residuals	0.2846	0.38	0.3594	82.6%

Baselines are fitted once, so no optimization-seed SD is attached. Residual samples are drawn from in-sample training errors and can understate uncertainty. Deterministic baselines have degenerate intervals. Complete results also include channel-sum CRPS, interval width, point MSE and coverage for every neural cell.

ETTh1: oil temperature

Head	White CRPS ↓	GP CRPS ↓	White energy ↓	GP energy ↓	White roughness	GP roughness
MLP	0.1422 ± 0.03	0.195 ± 0.017	0.1603 ± 0.032	0.2156 ± 0.015	2.059 ± 0.067	0.8708 ± 0.029
Whitened MLP	0.1284 ± 0.03	0.1321 ± 0.02	0.1471 ± 0.033	0.1477 ± 0.019	1.885 ± 0.045	1.363 ± 0.091
Balanced-loss MLP	0.1265 ± 0.012	0.7095 ± 0.19	0.1447 ± 0.013	0.7179 ± 0.18	1.928 ± 0.17	1.646 ± 0.096
Fixed-mask MLP	0.1409 ± 0.021	0.2006 ± 0.034	0.1587 ± 0.022	0.2266 ± 0.038	2.086 ± 0.07	1.56 ± 0.2
S4	0.1704 ± 0.015	0.1437 ± 0.02	0.1904 ± 0.014	0.1639 ± 0.021	1.78 ± 0.29	0.6806 ± 0.063
U-Net	0.2045 ± 0.04	0.2002 ± 0.028	0.2251 ± 0.043	0.2188 ± 0.029	1.735 ± 0.051	0.996 ± 0.011
L0-gated U-Net	0.2006 ± 0.054	0.2114 ± 0.01	0.2154 ± 0.054	0.2261 ± 0.0091	1.738 ± 0.26	0.9525 ± 0.14
Dense history AE	0.1628 ± 0.02	0.1985 ± 0.048	0.1833 ± 0.021	0.2221 ± 0.051	1.906 ± 0.039	0.9093 ± 0.084
TopK history SAE	0.1981 ± 0.029	0.2277 ± 0.023	0.2159 ± 0.031	0.251 ± 0.022	2.023 ± 0.038	0.8889 ± 0.079
Dense conv. history	0.1446 ± 0.019	0.197 ± 0.03	0.1637 ± 0.021	0.2211 ± 0.033	2.092 ± 0.11	0.8993 ± 0.038
Sparse conv. history	0.1482 ± 0.019	0.2224 ± 0.017	0.1672 ± 0.021	0.2452 ± 0.02	2.148 ± 0.1	0.8819 ± 0.0084
Dense velocity dictionary	0.1428 ± 0.017	0.1649 ± 0.057	0.1615 ± 0.018	0.1853 ± 0.062	2.036 ± 0.027	1.055 ± 0.031
TopK velocity dictionary	0.2218 ± 0.011	0.1794 ± 0.029	0.2576 ± 0.012	0.202 ± 0.028	9.035 ± 0.35	2.116 ± 0.29
Dense mixture	0.1339 ± 0.012	0.1966 ± 0.039	0.1526 ± 0.012	0.2192 ± 0.042	2.654 ± 0.11	0.8726 ± 0.027
Top-2 mixture	0.1303 ± 0.0086	0.1859 ± 0.016	0.1492 ± 0.0074	0.2061 ± 0.017	2.511 ± 0.077	0.8432 ± 0.027

Mean on the first line; ± sample SD across three optimization seeds on the second. All scores use train-standardized units. Compare methods within this dataset. White denotes temporally white noise with cross-channel covariance retained.

Simple forecasting controls

Baseline	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Persistence	0.1169	0.1427	0.02573	4.2%
Seasonal naive	0.1575	0.187	0.03989	2.3%
Ridge + residuals	0.09333	0.1104	0.02782	96.8%
Elastic Net + residuals	0.09194	0.1095	0.02684	97.0%

Baselines are fitted once, so no optimization-seed SD is attached. Residual samples are drawn from in-sample training errors and can understate uncertainty. Deterministic baselines have degenerate intervals. Complete results also include channel-sum CRPS, interval width, point MSE and coverage for every neural cell.

ETTh1: seven channels

Head	White CRPS ↓	GP CRPS ↓	White energy ↓	GP energy ↓	White roughness	GP roughness
MLP	0.3621 ± 0.0091	0.375 ± 0.016	0.4696 ± 0.0094	0.5069 ± 0.024	3.212 ± 0.044	0.6249 ± 0.051
Whitened MLP	0.5268 ± 0.013	0.5298 ± 0.01	0.6721 ± 0.017	0.669 ± 0.014	3.581 ± 0.012	0.7495 ± 0.029
Balanced-loss MLP	0.524 ± 0.027	0.574 ± 0.0026	0.6854 ± 0.026	0.735 ± 0.0024	3.346 ± 0.047	0.7185 ± 0.038
Fixed-mask MLP	0.3585 ± 0.017	0.3751 ± 0.022	0.4671 ± 0.022	0.5105 ± 0.022	2.869 ± 0.02	1.102 ± 0.083
S4	0.4045 ± 0.036	0.4033 ± 0.02	0.5483 ± 0.035	0.542 ± 0.031	1.471 ± 0.054	0.452 ± 0.025
U-Net	0.3825 ± 0.038	0.4382 ± 0.0097	0.5403 ± 0.045	0.6186 ± 0.025	1.111 ± 0.069	0.5263 ± 0.057
L0-gated U-Net	0.456 ± 0.026	0.5125 ± 0.031	0.5984 ± 0.026	0.6717 ± 0.032	1.791 ± 0.17	0.5298 ± 0.013
Dense history AE	0.4171 ± 0.0055	0.3908 ± 0.032	0.5342 ± 0.008	0.5273 ± 0.035	3.896 ± 0.06	0.577 ± 0.069
TopK history SAE	0.4635 ± 0.029	0.4364 ± 0.021	0.5907 ± 0.029	0.5935 ± 0.024	3.9 ± 0.077	0.5425 ± 0.056
Dense conv. history	0.3916 ± 0.0071	0.3862 ± 0.019	0.5037 ± 0.0079	0.5267 ± 0.011	3.115 ± 0.0031	0.5963 ± 0.011
Sparse conv. history	0.3959 ± 0.01	0.3863 ± 0.015	0.5078 ± 0.013	0.5299 ± 0.0039	3.107 ± 0.011	0.5943 ± 0.029
Dense velocity dictionary	0.4006 ± 0.0026	0.4116 ± 0.02	0.5197 ± 0.0068	0.5575 ± 0.0033	3.647 ± 0.028	0.8015 ± 0.024
TopK velocity dictionary	0.4813 ± 0.029	0.4789 ± 0.017	0.6073 ± 0.03	0.6182 ± 0.011	4.361 ± 0.023	1.208 ± 0.21
Dense mixture	0.377 ± 0.015	0.3566 ± 0.024	0.4876 ± 0.015	0.4903 ± 0.023	3.645 ± 0.01	0.6386 ± 0.09
Top-2 mixture	0.4046 ± 0.016	0.3401 ± 0.0033	0.5215 ± 0.02	0.4682 ± 0.0031	3.862 ± 0.023	0.6497 ± 0.026

Mean on the first line; ± sample SD across three optimization seeds on the second. All scores use train-standardized units. Compare methods within this dataset. White denotes temporally white noise with cross-channel covariance retained.

Simple forecasting controls

Baseline	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Persistence	0.5727	0.9214	0.9507	2.6%
Seasonal naive	0.3868	0.5966	0.4133	2.7%
Ridge + residuals	0.287	0.3984	0.3395	82.8%
Elastic Net + residuals	0.2977	0.4064	0.3529	82.5%

Baselines are fitted once, so no optimization-seed SD is attached. Residual samples are drawn from in-sample training errors and can understate uncertainty. Deterministic baselines have degenerate intervals. Complete results also include channel-sum CRPS, interval width, point MSE and coverage for every neural cell.

Exchange: eight channels

Head	White CRPS ↓	GP CRPS ↓	White energy ↓	GP energy ↓	White roughness	GP roughness
MLP	0.3137 ± 0.019	0.3852 ± 0.035	0.3819 ± 0.024	0.4451 ± 0.033	19.92 ± 0.79	1.725 ± 0.028
Whitened MLP	0.7071 ± 0.2	0.564 ± 0.012	0.8479 ± 0.24	0.6904 ± 0.0074	31.56 ± 1.2	3.963 ± 0.17
Balanced-loss MLP	1.114 ± 0.064	1.085 ± 0.00028	1.306 ± 0.068	1.269 ± 0.00045	27.71 ± 0.67	3.79 ± 0.048
Fixed-mask MLP	0.2837 ± 0.023	0.3011 ± 0.026	0.3472 ± 0.026	0.364 ± 0.029	20.21 ± 1.7	8.816 ± 1
S4	0.3354 ± 0.07	0.4255 ± 0.038	0.3934 ± 0.084	0.5048 ± 0.052	10.79 ± 2.8	1.741 ± 0.16
U-Net	0.5562 ± 0.023	0.5927 ± 0.067	0.668 ± 0.038	0.7105 ± 0.078	14.87 ± 2.4	2.98 ± 0.3
L0-gated U-Net	0.5495 ± 0.019	0.5675 ± 0.004	0.6555 ± 0.016	0.6782 ± 0.0033	16.16 ± 0.28	2.702 ± 0.51
Dense history AE	0.3439 ± 0.023	0.4485 ± 0.06	0.4191 ± 0.03	0.5287 ± 0.061	24.65 ± 0.87	1.785 ± 0.077
TopK history SAE	0.4243 ± 0.018	0.5304 ± 0.035	0.5173 ± 0.012	0.635 ± 0.03	24.29 ± 0.86	2.076 ± 0.2
Dense conv. history	0.3166 ± 0.058	0.475 ± 0.042	0.3879 ± 0.074	0.5686 ± 0.053	19.45 ± 0.44	1.959 ± 0.12
Sparse conv. history	0.2666 ± 0.042	0.478 ± 0.049	0.3245 ± 0.05	0.5718 ± 0.063	19.1 ± 0.38	1.96 ± 0.11
Dense velocity dictionary	0.3819 ± 0.052	0.4134 ± 0.035	0.4679 ± 0.063	0.4886 ± 0.031	22.97 ± 0.3	7.148 ± 0.96
TopK velocity dictionary	0.5018 ± 0.017	0.4868 ± 0.017	0.611 ± 0.019	0.5887 ± 0.012	36.06 ± 0.52	10.41 ± 1.3
Dense mixture	0.215 ± 0.0016	0.25 ± 0.034	0.2633 ± 0.0011	0.3065 ± 0.046	21.28 ± 0.3	2.532 ± 0.24
Top-2 mixture	0.1893 ± 0.014	0.2349 ± 0.025	0.2315 ± 0.017	0.2854 ± 0.031	21.51 ± 0.58	2.272 ± 0.054

Mean on the first line; ± sample SD across three optimization seeds on the second. All scores use train-standardized units. Compare methods within this dataset. White denotes temporally white noise with cross-channel covariance retained.

Simple forecasting controls

Baseline	CRPS ↓	Joint energy ↓	Mean MSE ↓	90% coverage
Persistence	0.08099	0.1124	0.01457	1.1%
Ridge + residuals	0.06778	0.09259	0.01922	81.5%
Elastic Net + residuals	0.06764	0.09144	0.0189	81.0%

Baselines are fitted once, so no optimization-seed SD is attached. Residual samples are drawn from in-sample training errors and can understate uncertainty. Deterministic baselines have degenerate intervals. Complete results also include channel-sum CRPS, interval width, point MSE and coverage for every neural cell.

Did the models actually become sparse?

Validation activity	GP (3C)	ETTh1 (1C)	ETTh1 (7C)	Exchange (8C)
Dense history AE	59.6-71.4%	50.1-56.2%	49.4-55.3%	46.2-52.5%
TopK history SAE	12.5-12.5%	12.5-12.5%	12.5-12.5%	12.5-12.5%
Dense conv. history	100.0-100.0%	100.0-100.0%	100.0-100.0%	100.0-100.0%
Sparse conv. history	88.8-94.8%	64.4-82.4%	92.1-95.0%	96.6-99.1%
Dense velocity dictionary	100.0-100.0%	100.0-100.0%	100.0-100.0%	100.0-100.0%
TopK velocity dictionary	12.5-12.5%	12.5-12.5%	12.5-12.5%	12.5-12.5%
L0 nonzero eval gates	100.0-100.0%	100.0-100.0%	100.0-100.0%	100.0-100.0%
L0 expected active gates	85.2-86.1%	84.4-85.3%	87.1-87.4%	84.7-85.9%
Dense mixture	100.0-100.0%	100.0-100.0%	100.0-100.0%	100.0-100.0%
Top-2 mixture	50.0-50.0%	50.0-50.0%	50.0-50.0%	50.0-50.0%

Ranges cover both sources and all three seeds on the full validation split. Code entries count nonzero coordinates; mixture entries count routed experts. All deterministic L0 gates remain nonzero in these runs: this is learned gating/attenuation, not successful channel pruning.

Sparse variants versus their paired dense controls

Sparse variant	Control	Lower CRPS	Range of CRPS differences
TopK history SAE	Dense history AE	0/8	+0.02918 to +0.08189
Sparse conv. history	Dense conv. history	1/8	-0.05005 to +0.02536
TopK velocity dictionary	Dense velocity dictionary	0/8	+0.01168 to +0.1198
L0-gated U-Net	U-Net	3/8	-0.02517 to +0.08025
Top-2 mixture	Dense mixture	6/8	-0.02571 to +0.02763

These counts summarize observed mean differences, not significant wins. Dense/TopK pairs hold width fixed. Sparse convolution changes both shrinkage and its L1 auxiliary penalty; L0 changes both stochastic training and gating regularization. The comparison does not isolate each ingredient separately.

Elastic Net feature selection

Dataset	Active history features	Alpha	L1 ratio
GP (3 channels)	25/96	0.01	1
ETTh1 (OT only)	29/48	0.01	1
ETTh1 (7 channels)	191/336	0.1	1
Exchange (8 channels)	70/256	0.01	1

The sparse neural implementations still execute dense PyTorch kernels. Activity reduction is not a measured MPS speedup or memory saving. Training times are recorded, but this suite is not a controlled runtime benchmark.

Correlated GP: three channels

Head	White PC1 MSE ↓	GP PC1 MSE ↓	White residual MSE ↓	GP residual MSE ↓
MLP	6.353 ± 0.054	15.95 ± 0.5	0.2022 ± 0.0043	0.2892 ± 0.0044
Whitened MLP	7.601 ± 0.15	14.39 ± 0.39	0.1888 ± 0.0028	0.2478 ± 0.0045
Balanced-loss MLP	9.942 ± 0.39	51.81 ± 0.012	0.2296 ± 0.0047	0.6001 ± 0.001
Fixed-mask MLP	6.278 ± 0.077	15.9 ± 0.26	0.2137 ± 0.0038	0.2946 ± 0.0033
S4	6.002 ± 0.23	16.17 ± 0.74	0.2001 ± 0.0025	0.3148 ± 0.0085
U-Net	5.516 ± 0.061	15.47 ± 0.45	0.1696 ± 0.004	0.2968 ± 0.011
L0-gated U-Net	6.492 ± 0.27	18.67 ± 1.3	0.5246 ± 0.13	0.4809 ± 0.093
Dense history AE	6.393 ± 0.06	17.25 ± 0.24	0.2244 ± 0.0061	0.3106 ± 0.0019
TopK history SAE	7.612 ± 0.26	20.48 ± 0.23	0.2428 ± 0.002	0.35 ± 0.0036
Dense conv. history	8.717 ± 0.097	21.43 ± 0.37	0.2461 ± 0.0096	0.3392 ± 0.0087
Sparse conv. history	8.725 ± 0.23	22.58 ± 0.75	0.2481 ± 0.011	0.3465 ± 0.0081
Dense velocity dictionary	6.474 ± 0.32	16.13 ± 0.67	0.2143 ± 0.01	0.2924 ± 0.0059
TopK velocity dictionary	7.166 ± 0.2	15.79 ± 0.16	0.6403 ± 0.0094	0.3545 ± 0.0022
Dense mixture	6.237 ± 0.19	15.62 ± 0.25	0.29 ± 0.0023	0.2974 ± 0.0025
Top-2 mixture	6.425 ± 0.33	15.47 ± 0.64	0.3043 ± 0.0098	0.2973 ± 0.0036

Mean ± sample SD across three seeds. Residual MSE is the average per-PC velocity error over PCs 3 onward in the joint F×C space. Outputs of the whitened MLP are mapped back before scoring. The original assignment metrics remain raw, unweighted errors; variance-normalized diagnostics are supplementary.

Train-fitted geometry and normalized residual diagnostic

Source	Validation velocity E2	Floored PCs	Raw MLP normalized	Whitened normalized	Balanced normalized
white	0.3955	0	0.135	0.1052	0.1365
gp	0.6783	30	0.2915	0.2098	0.3248

PCA variances are floored at $\max(10^{-6} \times \text{largest variance}, 10^{-8})$. Full-rank whitening rotates/scales the interpolant and predicts transformed velocity; balanced loss only reweights error in the same PCA basis. Their comparison tests the coordinate parameterization under the shared variance-weighted objective at this fixed optimization budget. Neither removes PCs.

Source-specific velocity labels and covariance spectra differ. A smaller velocity MSE under one source is not directly equivalent to a better forecast; read this page together with its CRPS, energy and roughness table. The original rank-two acceptance gate and two-sided 20% rule were not imposed on these broader real-data/multivariate experiments.

ETTh1: oil temperature

Head	White PC1 MSE ↓	GP PC1 MSE ↓	White residual MSE ↓	GP residual MSE ↓
MLP	1.598 ± 0.36	10.29 ± 0.95	0.1175 ± 0.0051	0.03569 ± 0.0036
Whitened MLP	1.589 ± 0.37	6.547 ± 1.1	0.1115 ± 0.0086	0.03052 ± 0.0018
Balanced-loss MLP	1.642 ± 0.12	45.31 ± 12	0.1141 ± 0.0077	0.03474 ± 0.0023
Fixed-mask MLP	1.507 ± 0.29	11.46 ± 4.6	0.1168 ± 0.0069	0.0423 ± 0.0029
S4	2.555 ± 0.39	6.123 ± 1.5	0.1167 ± 0.0035	0.03983 ± 0.0025
U-Net	3.329 ± 0.79	11.06 ± 3.3	0.1125 ± 0.0034	0.03947 ± 0.0022
L0-gated U-Net	3.161 ± 1.3	10.04 ± 1.7	0.1248 ± 0.00055	0.04322 ± 0.005
Dense history AE	1.726 ± 0.31	10.58 ± 4.9	0.1206 ± 0.0073	0.0381 ± 0.0014
TopK history SAE	2.553 ± 0.51	12.72 ± 2.8	0.1156 ± 0.0054	0.03775 ± 0.0037
Dense conv. history	1.545 ± 0.22	9.55 ± 2.1	0.1183 ± 0.0037	0.03933 ± 0.0056
Sparse conv. history	1.639 ± 0.27	12.23 ± 2.2	0.1205 ± 0.0092	0.03828 ± 0.0032
Dense velocity dictionary	1.564 ± 0.18	8.756 ± 4.2	0.1202 ± 0.0045	0.03658 ± 0.0012
TopK velocity dictionary	3.165 ± 0.47	10.03 ± 3.8	0.3725 ± 0.0082	0.05176 ± 0.0035
Dense mixture	1.507 ± 0.23	9.798 ± 2.7	0.1393 ± 0.0088	0.03642 ± 0.0039
Top-2 mixture	1.498 ± 0.16	8.801 ± 1	0.1389 ± 0.008	0.03736 ± 0.004

Mean ± sample SD across three seeds. Residual MSE is the average per-PC velocity error over PCs 3 onward in the joint F×C space. Outputs of the whitened MLP are mapped back before scoring. The original assignment metrics remain raw, unweighted errors; variance-normalized diagnostics are supplementary.

Train-fitted geometry and normalized residual diagnostic

Source	Validation velocity E2	Floored PCs	Raw MLP normalized	Whitened normalized	Balanced normalized
white	0.2652	0	0.1109	0.105	0.1073
gp	0.9076	0	0.3758	0.0838	0.1301

PCA variances are floored at $\max(10^{-6} \times \text{largest variance}, 10^{-8})$. Full-rank whitening rotates/scales the interpolant and predicts transformed velocity; balanced loss only reweights error in the same PCA basis. Their comparison tests the coordinate parameterization under the shared variance-weighted objective at this fixed optimization budget. Neither removes PCs.

Source-specific velocity labels and covariance spectra differ. A smaller velocity MSE under one source is not directly equivalent to a better forecast; read this page together with its CRPS, energy and roughness table. The original rank-two acceptance gate and two-sided 20% rule were not imposed on these broader real-data/multivariate experiments.

ETTh1: seven channels

Head	White PC1 MSE ↓	GP PC1 MSE ↓	White residual MSE ↓	GP residual MSE ↓
MLP	5.643 ± 0.67	26.06 ± 5.6	1.036 ± 0.013	0.8011 ± 0.032
Whitened MLP	16.86 ± 3.7	67.19 ± 7.6	1.568 ± 0.016	1.317 ± 0.12
Balanced-loss MLP	8.108 ± 0.79	49.66 ± 0.038	1.642 ± 0.057	1.501 ± 0.0051
Fixed-mask MLP	5.901 ± 1.8	21.53 ± 0.63	1.002 ± 0.017	0.7916 ± 0.075
S4	12.02 ± 5.6	34.86 ± 9.2	0.832 ± 0.037	0.7579 ± 0.048
U-Net	6.314 ± 0.76	23.7 ± 4.7	0.8316 ± 0.051	0.9026 ± 0.11
L0-gated U-Net	7.706 ± 2.6	20.9 ± 2.3	0.9411 ± 0.07	0.9061 ± 0.02
Dense history AE	7.532 ± 1.1	23.8 ± 3.3	1.278 ± 0.0073	0.8757 ± 0.11
TopK history SAE	13.12 ± 5	30.01 ± 3.6	1.315 ± 0.1	0.9493 ± 0.052
Dense conv. history	4.98 ± 0.072	20.13 ± 1.3	1.047 ± 0.0088	0.7475 ± 0.028
Sparse conv. history	5.679 ± 0.62	20.23 ± 1.6	1.053 ± 0.019	0.7511 ± 0.019
Dense velocity dictionary	6.573 ± 0.93	34.12 ± 9.1	1.223 ± 0.018	0.9099 ± 0.11
TopK velocity dictionary	16.07 ± 6.6	34.74 ± 5.3	1.601 ± 0.052	1.184 ± 0.052
Dense mixture	6.431 ± 0.74	21.99 ± 4.6	1.125 ± 0.012	0.7444 ± 0.095
Top-2 mixture	8.118 ± 2.1	22.9 ± 3.5	1.258 ± 0.087	0.6501 ± 0.013

Mean ± sample SD across three seeds. Residual MSE is the average per-PC velocity error over PCs 3 onward in the joint F×C space. Outputs of the whitened MLP are mapped back before scoring. The original assignment metrics remain raw, unweighted errors; variance-normalized diagnostics are supplementary.

Train-fitted geometry and normalized residual diagnostic

Source	Validation velocity E2	Floored PCs	Raw MLP normalized	Whitened normalized	Balanced normalized
white	0.2099	0	0.9527	0.9215	0.9452
gp	0.3735	0	1.129	0.9254	0.9855

PCA variances are floored at $\max(10^{-6} \times \text{largest variance}, 10^{-8})$. Full-rank whitening rotates/scales the interpolant and predicts transformed velocity; balanced loss only reweights error in the same PCA basis. Their comparison tests the coordinate parameterization under the shared variance-weighted objective at this fixed optimization budget. Neither removes PCs.

Source-specific velocity labels and covariance spectra differ. A smaller velocity MSE under one source is not directly equivalent to a better forecast; read this page together with its CRPS, energy and roughness table. The original rank-two acceptance gate and two-sided 20% rule were not imposed on these broader real-data/multivariate experiments.

Exchange: eight channels

Head	White PC1 MSE ↓	GP PC1 MSE ↓	White residual MSE ↓	GP residual MSE ↓
MLP	22.48 ± 8.8	32.81 ± 7.3	0.5539 ± 0.049	0.8828 ± 0.15
Whitened MLP	60.59 ± 56	148.5 ± 89	1.241 ± 0.081	1.222 ± 0.15
Balanced-loss MLP	267.7 ± 1.5	347.8 ± 0.055	1.111 ± 0.2	1.329 ± 0.0013
Fixed-mask MLP	20.09 ± 14	32.45 ± 13	0.5439 ± 0.072	0.6289 ± 0.049
S4	27.25 ± 17	30.17 ± 8.9	0.4728 ± 0.2	0.7245 ± 0.07
U-Net	30.2 ± 13	79.95 ± 45	0.916 ± 0.037	1.115 ± 0.055
L0-gated U-Net	14.4 ± 4	54.62 ± 7.9	0.9575 ± 0.016	0.9505 ± 0.021
Dense history AE	10.02 ± 2.6	32.62 ± 10	0.7649 ± 0.053	1.182 ± 0.15
TopK history SAE	15.57 ± 2.2	33.42 ± 13	0.9381 ± 0.079	1.222 ± 0.12
Dense conv. history	9.719 ± 1.9	24.95 ± 7	0.5565 ± 0.074	0.9772 ± 0.07
Sparse conv. history	12.32 ± 5.4	22.04 ± 6.8	0.4695 ± 0.064	0.9648 ± 0.083
Dense velocity dictionary	20.16 ± 8.5	31.73 ± 7.3	0.7214 ± 0.056	1.061 ± 0.11
TopK velocity dictionary	16.61 ± 1.2	23.33 ± 3.7	1.505 ± 0.077	1.024 ± 0.1
Dense mixture	8.997 ± 3.5	25.58 ± 8	0.4758 ± 0.033	0.4804 ± 0.061
Top-2 mixture	6.43 ± 2.5	20.73 ± 5.6	0.4647 ± 0.035	0.4991 ± 0.085

Mean ± sample SD across three seeds. Residual MSE is the average per-PC velocity error over PCs 3 onward in the joint $F \times C$ space. Outputs of the whitened MLP are mapped back before scoring. The original assignment metrics remain raw, unweighted errors; variance-normalized diagnostics are supplementary.

Train-fitted geometry and normalized residual diagnostic

Source	Validation velocity E2	Floored PCs	Raw MLP normalized	Whitened normalized	Balanced normalized
white	0.1529	0	0.8688	0.9172	0.873
gp	0.5952	1	2.516	1.979	2.558

PCA variances are floored at $\max(10^{-6} \times \text{largest variance}, 10^{-8})$. Full-rank whitening rotates/scales the interpolant and predicts transformed velocity; balanced loss only reweights error in the same PCA basis. Their comparison tests the coordinate parameterization under the shared variance-weighted objective at this fixed optimization budget. Neither removes PCs.

Source-specific velocity labels and covariance spectra differ. A smaller velocity MSE under one source is not directly equivalent to a better forecast; read this page together with its CRPS, energy and roughness table. The original rank-two acceptance gate and two-sided 20% rule were not imposed on these broader real-data/multivariate experiments.

Separate sparsity from a rank restriction

The main suite uses 64 learned velocity templates (dictionary atoms). That can span the 48-dimensional GP future and 24-dimensional OT future, but it cannot span a 168-dimensional ETTh1 joint future or a 128-dimensional Exchange future. This limitation applies to both dense and TopK coefficients.

Why rank matters. With velocity $v = D a$, the ODE can change only directions in the column span of D . Every component orthogonal to that span remains equal to its initial source value. A low-rank velocity dictionary does not make the complete generated endpoint low rank: unused source variation survives.

Dataset	Learned templates	Learned rank range	Unchangeable directions
ETTh1 (7 channels)	64	64-64	104-104
ETTh1 (7 channels)	192	168-168	0-0
Exchange (8 channels)	64	64-64	64-64
Exchange (8 channels)	192	128-128	0-0

Ranges cover both sources, both coefficient heads and all three seeds. Learned rank is checked from each saved dictionary, not inferred from its number of columns. The independent audit also checks the generated endpoint orthogonal component against the exact initial source samples.

A declared 24-fit structural control

We added 192-template dense and TopK heads on ETTh1 (7 channels) and Exchange (8 channels), with both sources and three seeds. No main result was overwritten. All data, optimizer settings, training steps, validation selection, forecast contexts and solver settings match the 64-template experiment. TopK still keeps at most eight coefficients per example.

Dataset	Templates	Predictor width	Stored parameters	TopK budget
ETTh1 (7 channels)	64	40	64,696	8/64
ETTh1 (7 channels)	192	26	64,246	8/192
Exchange (8 channels)	64	43	63,339	8/64
Exchange (8 channels)	192	32	63,520	8/192

The approximately 64k parameter budget makes the coefficient predictor narrower when the dictionary grows. Thus 192-versus-64 changes capacity allocation as well as removing the template-count rank cap. Within each size, dense and TopK use identical widths and parameter counts. This is an exploratory follow-up designed after inspecting the main architecture, not a preregistered pure rank intervention.

A TopK head selects eight templates at each state, but its active set can change along the trajectory. Its field is therefore not permanently rank eight. Conversely, a full-rank 192-template dictionary does not guarantee that the coefficient model actually uses most templates.

Measured 64-versus-192 comparisons

ETTh1 (7 channels)

Templates / coefficients	Source	CRPS ↓	Joint energy ↓	Roughness ratio
64 / dense	white	0.4006 ± 0.0026	0.5197 ± 0.0068	3.647 ± 0.028
64 / dense	gp	0.4116 ± 0.02	0.5575 ± 0.0033	0.8015 ± 0.024
64 / TopK8	white	0.4813 ± 0.029	0.6073 ± 0.03	4.361 ± 0.023
64 / TopK8	gp	0.4789 ± 0.017	0.6182 ± 0.011	1.208 ± 0.21
192 / dense	white	0.3868 ± 0.01	0.4994 ± 0.011	3.772 ± 0.034
192 / dense	gp	0.3967 ± 0.016	0.5382 ± 0.015	0.6395 ± 0.024
192 / TopK8	white	0.4658 ± 0.011	0.5914 ± 0.015	4.418 ± 0.013
192 / TopK8	gp	0.4966 ± 0.014	0.6336 ± 0.021	1.002 ± 0.14

Exchange (8 channels)

Templates / coefficients	Source	CRPS ↓	Joint energy ↓	Roughness ratio
64 / dense	white	0.3819 ± 0.052	0.4679 ± 0.063	22.97 ± 0.3
64 / dense	gp	0.4134 ± 0.035	0.4886 ± 0.031	7.148 ± 0.96
64 / TopK8	white	0.5018 ± 0.017	0.611 ± 0.019	36.06 ± 0.52
64 / TopK8	gp	0.4868 ± 0.017	0.5887 ± 0.012	10.41 ± 1.3
192 / dense	white	0.3542 ± 0.064	0.4352 ± 0.071	24.58 ± 0.42
192 / dense	gp	0.4112 ± 0.048	0.4937 ± 0.056	3.093 ± 1.2
192 / TopK8	white	0.4967 ± 0.0096	0.6056 ± 0.011	37.25 ± 0.72
192 / TopK8	gp	0.4831 ± 0.022	0.583 ± 0.026	7.607 ± 1.6

The larger vocabulary does not rescue TopK at this budget. TopK has higher mean CRPS than its dense control in 8/8 dataset/source/size comparisons. Mean $\pm$ sample SD uses three seeds; all 48 runs are retained. Paired seed differences and context-block intervals are supplied separately.

Dataset	192 lowers CRPS	CRPS change range	Templates visited
ETTh1 (7 channels)	3/4	-0.0155 to +0.01772	20-28 / 192
Exchange (8 channels)	4/4	-0.02774 to -0.002264	12-20 / 192

Template usage is the union over the complete validation split, separately for each fitted TopK model. It can be much smaller than the dictionary despite full learned matrix rank. Improved span and actual template utilization are separate questions; these descriptive counts do not establish a universal benefit of a larger or sparser dictionary.

What is checked, and what remains uncertain

Finite training budget. 125/264 new runs select the final checked step. The separate 24-fit full-rank dictionary follow-up is reported on the preceding pages. This flags possible optimization limits without proving convergence or failure. The suite uses one learning rate and fixed auxiliary-loss strengths; a weak sparse result does not establish that the method family cannot work.

Euler 64 to 128, new heads	Max endpoint RMSE	Max CRPS change
GP (3 channels)	0.007077	0.0005976
ETTh1 (OT only)	0.006387	0.0006506
ETTh1 (7 channels)	0.007959	0.0004616
Exchange (8 channels)	0.00844	0.001497

Solver checks use seed 31, the first eight selected contexts and all 11 new heads with both sources. They are a limited sensitivity check, not proof of convergence on every forecast.

Independent numerical and data checks

Independent audits cover all 264 main and 24 supplementary fits and rescore all 96 frozen controls. The checks reconstruct training scalars, splits, source covariance and PCA; verify artifact hashes and checkpoint selection; replay full validation and test velocities on CPU; and recompute proper scores, roughness and saved doubled-Euler arrays. They also reconstruct Elastic Net predictions and training residuals. This exporter separately checked 2,040 aggregate mean/SD pairs against saved run records.

The complete 122-test suite passed on MPS. All 36 actual-MPS model checks passed: nine sparse/dense variants at 1, 3, 7 and 8 channels. Maximum CPU-MPS difference was 2.24e-07 in forward output, 1.04e-07 in gradients including auxiliary terms, and 1.07e-06 after an AdamW update. Sparse supports agreed. Stochastic gates have private checkpointed RNG; forward calls leave training-data RNG untouched.

Data provenance and scientific limits

ETTh1 uses 48 observed and 24 future hours, with 12/4/4-month row splits. Exchange uses 32 observed and 16 future steps, with a 70/10/20 chronological split. The synthetic problem uses 32 to 16 steps and three correlated channels. These protocols differ from the original papers. Windows and train-standardized scales match the frozen extension controls.

Context bootstrap intervals are exploratory: iid resampling for independent GP contexts, circular blocks of four and eight for real contexts. They condition on the chosen periods and trained seeds; block sensitivity and multiple comparisons limit generalization. A model can have favorable marginal CRPS while missing cross-channel structure, calibration or low-variance components.

Primary methods and reproducibility

TopK SAE and its official code; CRsAE author implementation; hard-concrete L0 regularization; Time-MoE; MultiTask Elastic Net documentation. These motivate small independent adaptations; this report does not claim to reproduce their published results.

Run `scripts/run_sparse.sh`, or reproduce its stages with `fm_stress.sparse_experiment`, `fm_stress.sparse_baselines_qr`, `scripts/check_sparse_extension.py`, `fm_stress.analyze_sparse` and this builder. Configuration: `configs/sparse_extension.json`. Saved artifacts: `results/sparse-extension`; complete tables and paired contrasts: `output/sparse-analysis`. Model adaptations are documented in `docs/sparse_model_design.md`.